

EC-350 AI and Decision Support Systems

Week 10 K-Nearest Neighbour Classifier

Dr. Arslan Shaukat



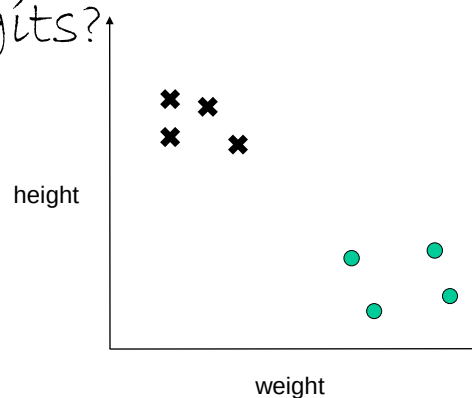
Acknowledgements: Lecture slides material from
Duda, Hart and Stork, Dr. Gavin Brown

1

Our Problem

Now – how is this problem like
recognising handwritten
digits?

7210414959
0690159784
9665407401
3134727121
1742351244



08/12/2022

EC-350 AI and DSS

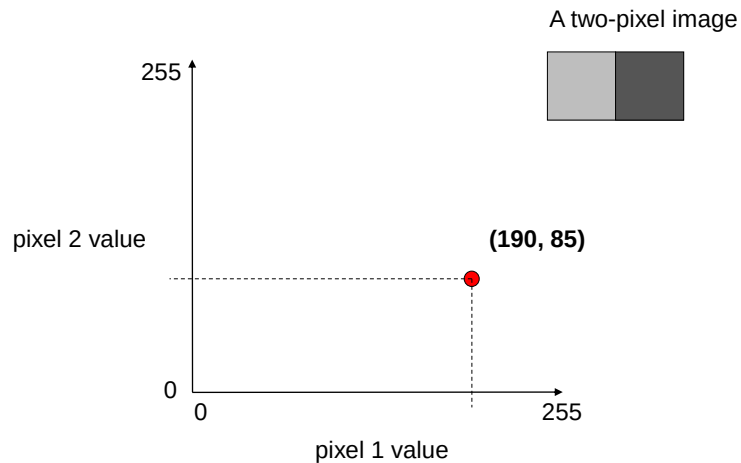
EME (NUST)

2

2

Recognising Handwritten Digits

Let's say the axes now represent **pixel values**.



08/12/2022

EC-350 AI and DSS

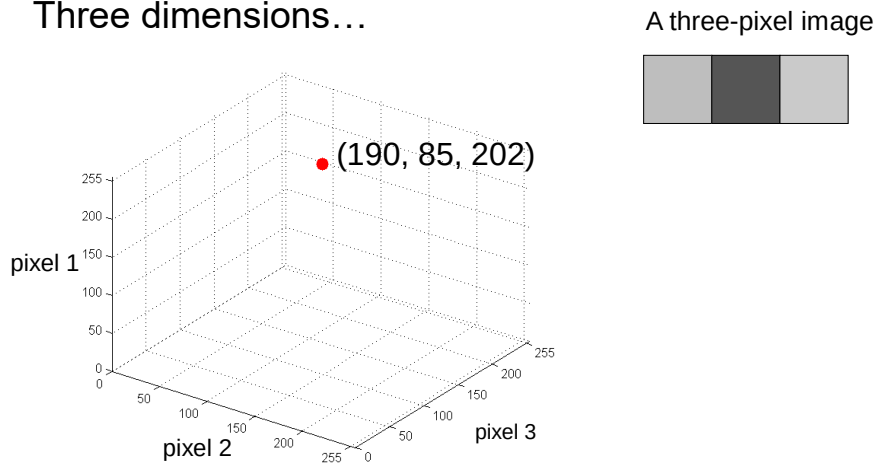
EME (NUST)

3

3

Recognising Handwritten Digits

Three dimensions...



This 3-pixel image is represented by a **SINGLE** point in a 3-D space.

08/12/2022

EC-350 AI and DSS

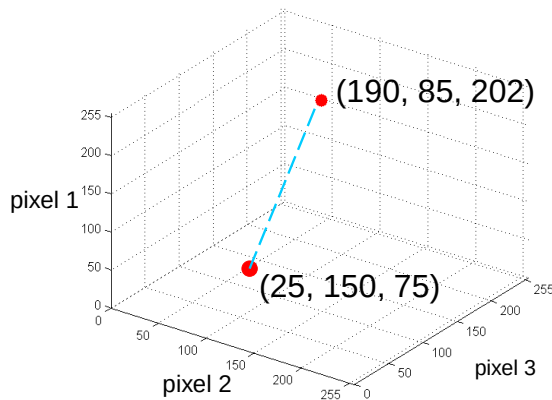
EME (NUST)

4

4

Recognising Handwritten Digits

Distances between images



A three-pixel image



Another 3-pixel image



Straight line distance
between them

08/12/2022

EC-350 AI and DSS

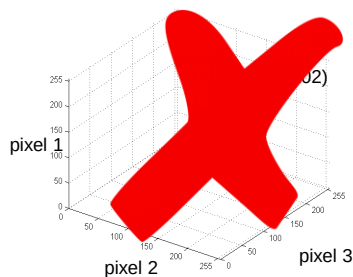
EME (NUST)

5

5

Recognising Handwritten Digits

Four dimensions? Five? Six-dimensional spaces?



A three-pixel image



A four-pixel image.



A five-pixel image



08/12/2022

EC-350 AI and DSS

EME (NUST)

6

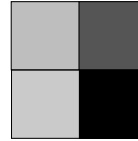
6

Recognising Handwritten Digits

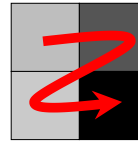
A four-pixel image.



A different four-pixel image.



(190, 85, 202, 10)
Same 4-dimensional vector!



Assuming we read pixels in a systematic manner, we can now represent any image as a single point in a high dimensional space.

08/12/2022

EC-350 AI and DSS

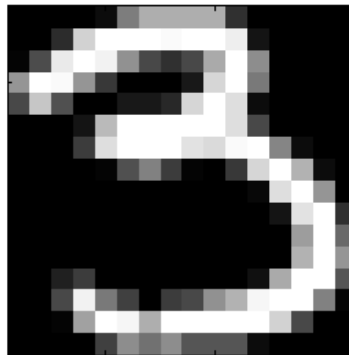
EME (NUST)

7

7

Recognising Handwritten Digits

16 x 16 image.... How many dimensions?



08/12/2022

EC-350 AI and DSS

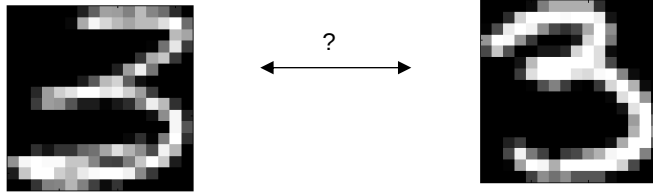
EME (NUST)

8

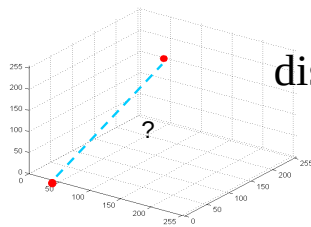
8

Recognising Handwritten Digits

Distances between digits....



Straight-line distance in N-dimensional space?



$$\text{distance}(x, x') = \sqrt{\sum_{i=1}^{i=256} (x_i - x'_i)^2}$$

08/12/2022

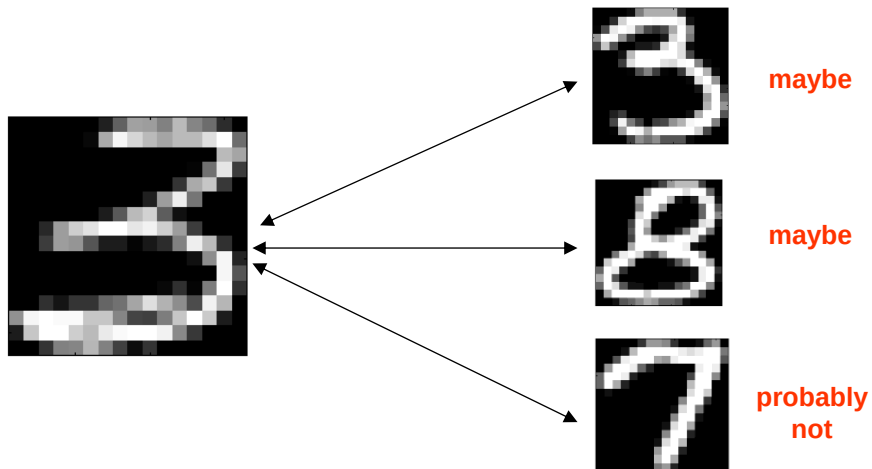
EC-350 AI and DSS

EME (NUST)

9

9

Recognising Handwritten Digits



Which is **closest neighbour** in N-dimensions?

08/12/2022

EC-350 AI and DSS

EME (NUST)

10

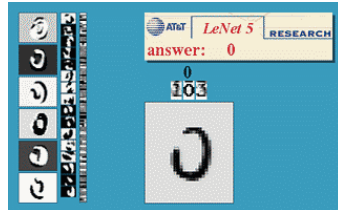
10

Recognising Handwritten Digits

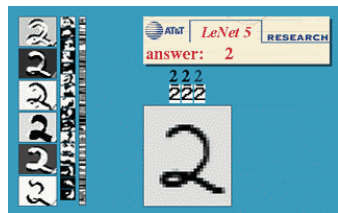


AT&T Research Labs

The USPS ZIP code reader – **learnt** from examples.



FAST recognition



NOISE resistant, can **generalise** to future unseen patterns

08/12/2022

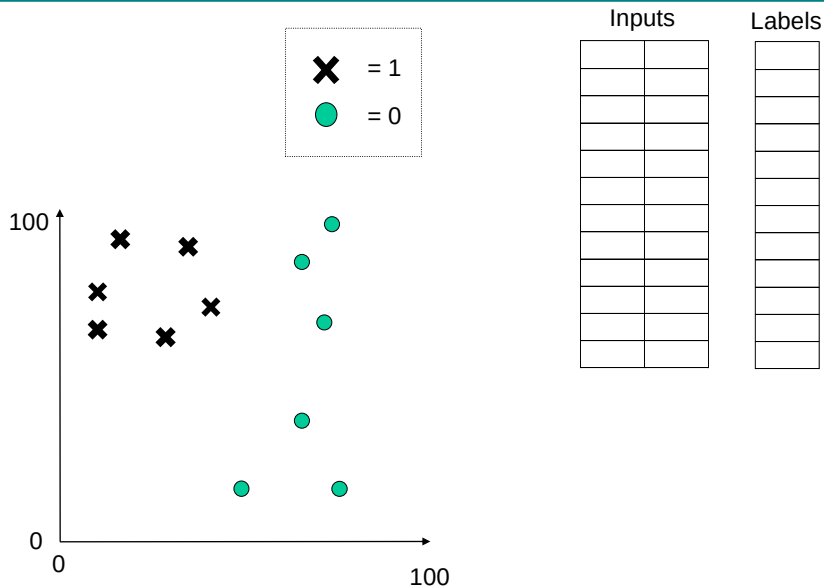
EC-350 AI and DSS

EME (NUST)

11

11

Training and Testing Data



08/12/2022

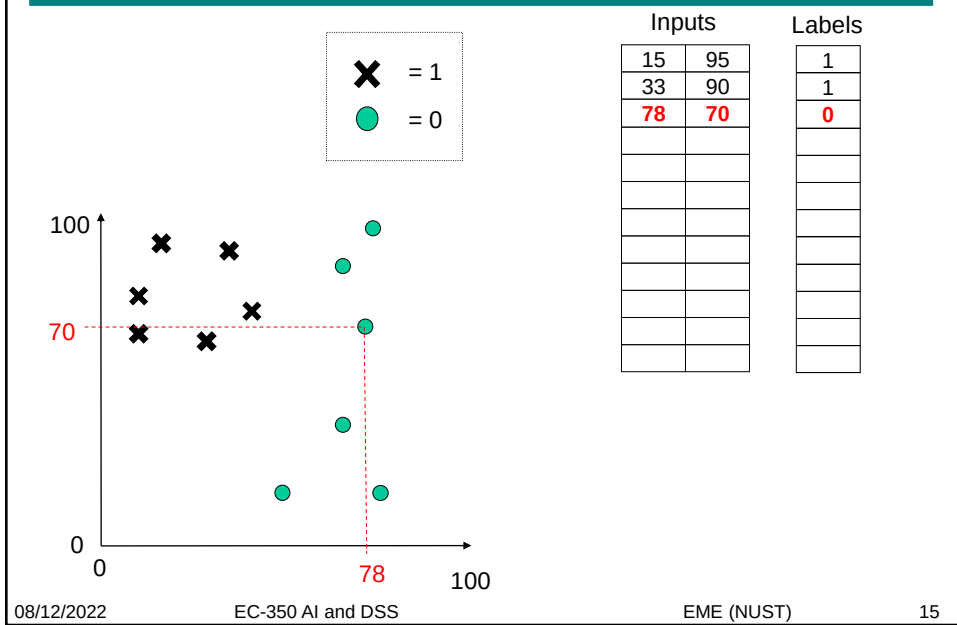
EC-350 AI and DSS

EME (NUST)

12

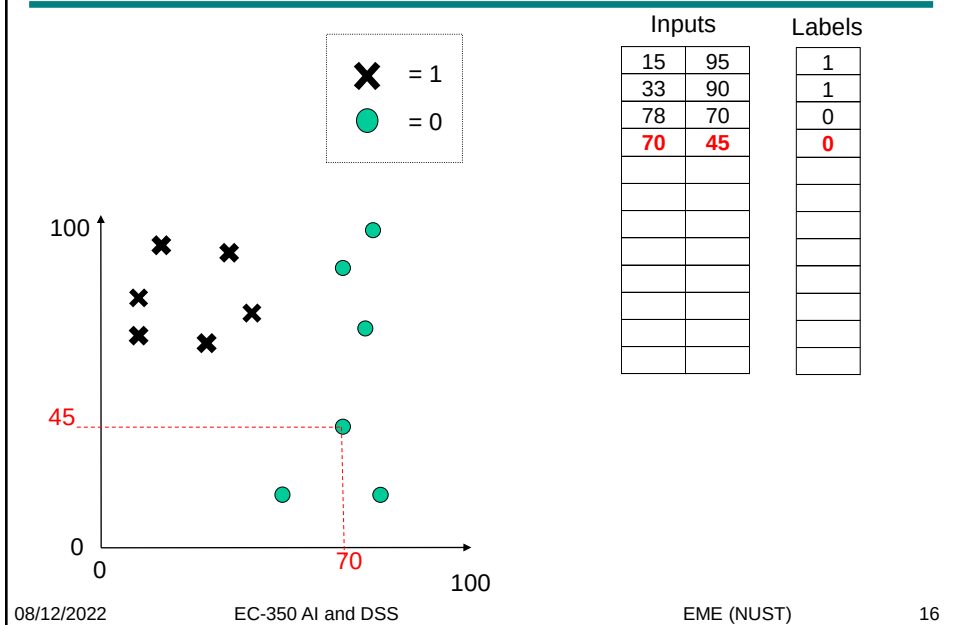
12

Training and Testing Data



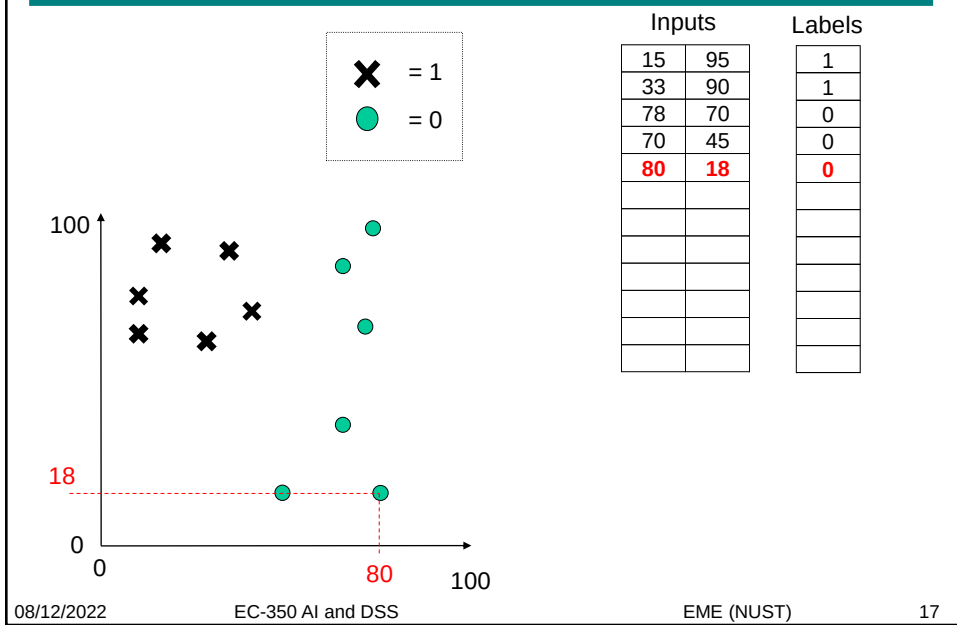
15

Training and Testing Data



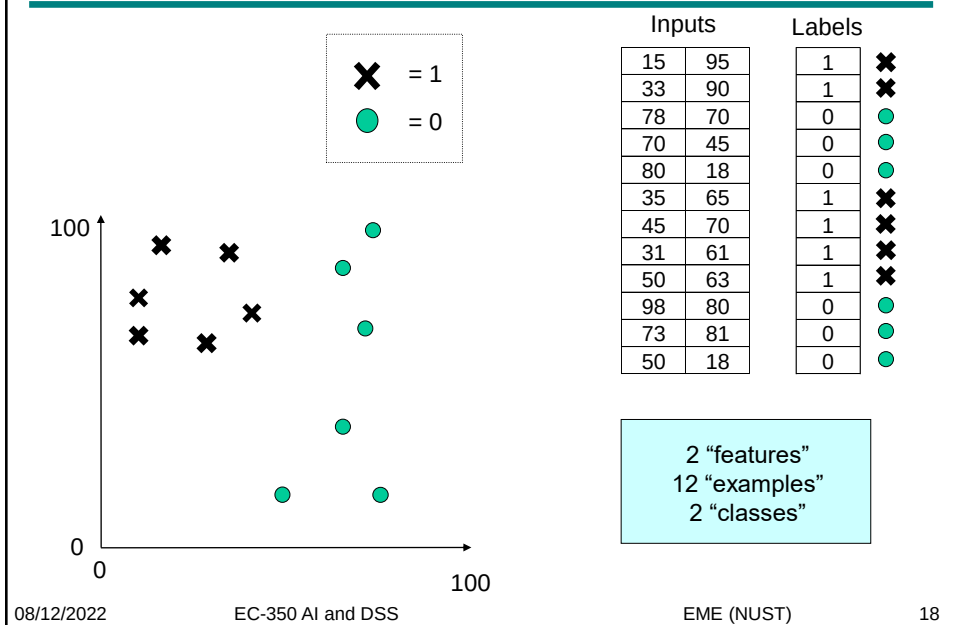
16

Training and Testing Data



17

Training and Testing Data



18

Training and Testing Data

Dataset

Inputs		Labels
15	95	1
33	90	1
78	70	0
70	45	0
80	18	0
35	65	1
45	70	1
31	61	1
50	63	1
98	80	0
73	81	0
50	18	0

08/12/2022

EC-350 AI and DSS

EME (NUST)

19

19

Training and Testing Data

Inputs		Labels				
15	95	1	 50:50 split	15	95	1
33	90	1		33	90	1
78	70	0		78	70	0
70	45	0		70	45	0
80	18	0		80	18	0
35	65	1		35	65	1
45	70	1				
31	61	1				
50	63	1				
98	80	0				
73	81	0		45	70	1
50	18	0		31	61	1
				50	63	1
				98	80	0
				73	81	0
				50	18	0

08/12/2022

EC-350 AI and DSS

EME (NUST)

20

20

Training and Testing Data

If too small... you'll make many mistakes on the testing set.

*Needs to be big.
Bigger is better.*

15	95	1
33	90	1
78	70	0
70	45	0
80	18	0
35	65	1

Training set

Train a K-NN on this...

45	70	1
31	61	1
50	63	1
98	80	0
73	81	0
50	18	0

Testing set

... then, test it on this!

"simulates" what it might be like to see new data in the future

08/12/2022

EC-350 AI and DSS

EME (NUST)

21

21

Training and Testing Data

Training set

Build a k-NN using training set

Testing set

Percentage of incorrect predictions is called the "error"

e.g. "Training" error
e.g. "Testing" error

How many incorrect predictions on testing set?

08/12/2022

EC-350 AI and DSS

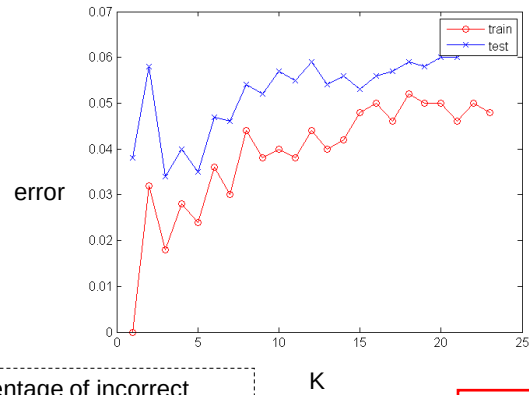
EME (NUST)

22

22

Classifying '3' versus '8' Digits

Plotting error as a function of 'K' (as in the K-NN)



Percentage of incorrect predictions is called the "error"

e.g. "Training" error
e.g. "Testing" error

K

Training data can behave very differently to testing data!

08/12/2022

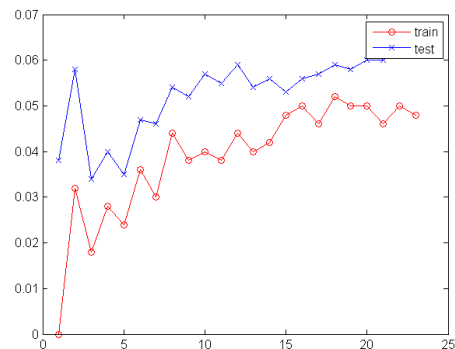
EC-350 AI and DSS

EME (NUST)

23

23

Classifying '3' versus '8' Digits



Best testing error at $K=3$, about 3.2%.

Is this "good"?

08/12/2022

EC-350 AI and DSS

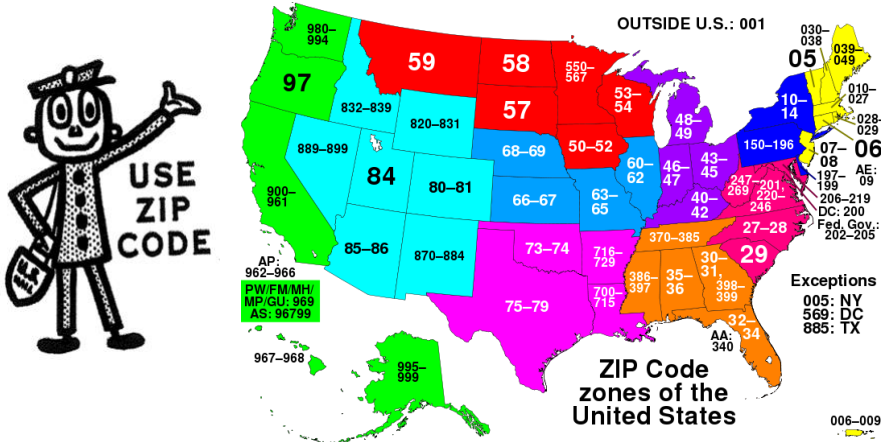
EME (NUST)

24

24

Classifying '3' versus '8' Digits

Zip-codes: "8" is very common on the West Coast, while "3" is rare.
Making a mistake will mean your Florida post ends up in Las Vegas!



08/12/2022

EC-350 AI and DSS

EME (NUST)

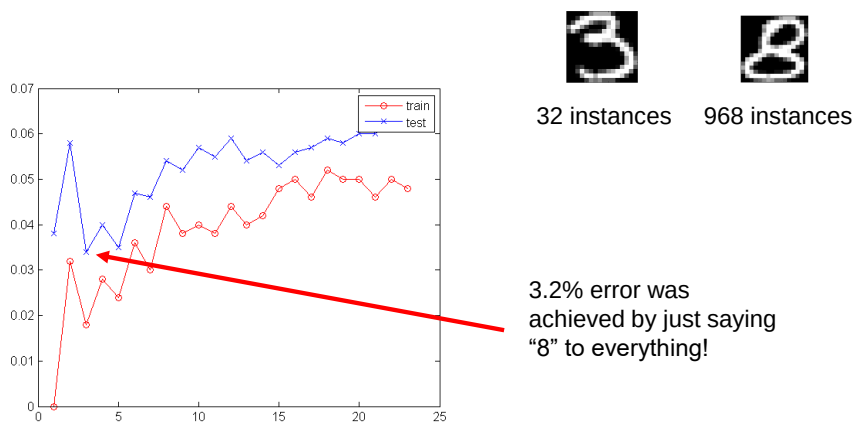
25

25

Classifying '3' versus '8' Digits

Sometimes, classes are rare, so your learner will not see many of them.

What if, in testing phase you saw 1,000 digits



08/12/2022

EC-350 AI and DSS

EME (NUST)

26

26

Classifying '3' versus '8' Digits

Solution?



32 instances



968 instances

0% correct

100% correct

Measure accuracy on each class separately.

08/12/2022

EC-350 AI and DSS

EME (NUST)

27

27

R.O.C. Analysis



32 instances 968 instances

A statistical framework.

Receiver Operator Characteristics

Developed in WW-2 to assess radar operators.

"How good is the radar operator at spotting incoming bombers?"



False positives

– i.e. falsely predicting a bombing raid

False negatives

*– i.e. missing an incoming bomber
(VERY BAD!)*

08/12/2022

EC-350 AI and DSS

EME (NUST)

28

28

R.O.C. Analysis

The “3” digits are like the bombers.
Rare events but costly if we misclassify!

False positives – i.e. falsely predicting an event
False negatives – i.e. missing an incoming event



Similarly, we have “true positives” and “true negatives”

		Prediction	
		0	1
Truth	0	TN	FP
	1	FN	TP

08/12/2022

EC-350 AI and DSS

EME (NUST)

29

29

Building a “Confusion Matrix”

		Prediction	
		0	1
Truth	0	TN	FP
	1	FN	TP

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

... chances of spotting a “3” when presented with one
(i.e. accuracy on class “3”)

$$\text{Specificity} = \frac{TN}{TN+FP}$$

... chances of spotting an 8 when presented with one
(i.e. accuracy on class “8”)

08/12/2022

EC-350 AI and DSS

EME (NUST)

30

30

R.O.C. Analysis

$$\text{Sensitivity} = \frac{TP}{TP+FN} = ?$$

$$\text{Specificity} = \frac{TN}{TN+FP} = ?$$

		Prediction	
		0	1
Truth	0	60	30
	1	80	20

TN	FP
FN	TP

60+30 = 90 examples in the dataset were class 0

80+20 = 100 examples in the dataset were class 1

90+100 = 190 examples in the data overall

08/12/2022

EC-350 AI and DSS

EME (NUST)

31

31

Assignment # 3

- Use of K-NN classifier
- You will give the printed report including
 - Code
 - Results (Figures, Graphs, Tables etc.)
 - Your description (Max. 1 page)
- Deadline for submission
 - After 2 weeks

08/12/2022

EC-350 AI and DSS

EME (NUST)

32

32

EC-350 AI and Decision Support Systems

Bayesian Decision Theory



Acknowledgements: Lecture slides material from
Duda, Hart and Stork, Dr. Gavin Brown

33

Bayes Decision Theory

- Statistically best classifier
- Uses a probabilistic approach
- The theory assumes:
 - *Decision problem can be posed in probabilistic terms*
 - *All relevant probability values are known (in practice this is not true)*

08/12/2022

EC-350 AI and DSS

EME (NUST)

34

34

Introduction

- The sea bass/salmon example
 - *State of nature*
 - *a priori (prior) probability*
 - *State of nature is unpredictable, so it is a random variable*
 - *The catch of salmon and sea bass is equiprobable*
 - $P(\omega_1) = P(\omega_2)$ (uniform priors)
 - $P(\omega_1) + P(\omega_2) = 1$
 - *Prior prob. reflects our prior knowledge about how likely we are to observe a sea bass or salmon*
 - *These probabilities may depend on time of the year or the fishing area!*

08/12/2022

EC-350 AI and DSS

EME (NUST)

35

35

Decision Rule

- Decision rule with only the prior information
 - *Decide ω_1 if $P(\omega_1) > P(\omega_2)$, otherwise decide ω_2*
- Suppose now we have a measurement or feature on the state of nature - say the fish lightness value
- Use of the class –conditional probability density $P(x | \omega_1)$ and $P(x | \omega_2)$ describe the difference in lightness feature between populations of sea bass and salmon

08/12/2022

EC-350 AI and DSS

EME (NUST)

36

36

Class Conditional Probabilities

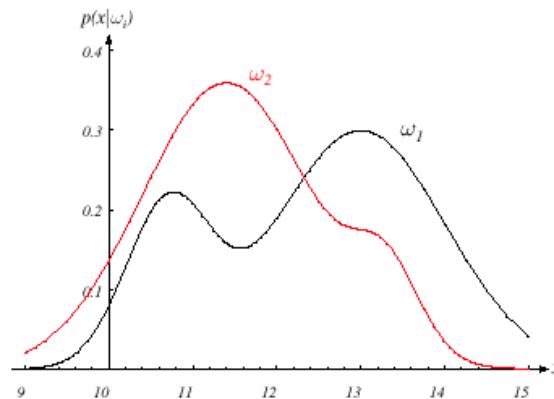


FIGURE 2.1. Hypothetical class-conditional probability density functions show the probability density of measuring a particular feature value x given the pattern is in category ω_i . If x represents the lightness of a fish, the two curves might describe the difference in lightness of populations of two types of fish. Density functions are normalized, and thus the area under each curve is 1.0. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons,

08/12/2022

EC-350 AI and DSS

EME (NUST)

37

37

Class Conditional Probabilities

- Decision rule
 - Assign input pattern x to class ω_1 if

$$P(x | \omega_1) > P(x | \omega_2), \text{ otherwise } \omega_2$$
- Suppose, we know prior and conditional probabilities and that we have observed the value of the feature x
 - How would you decide on the “state of nature”—type of fish, based on this info?
 - Bayes theory allows us to compute the posterior probabilities from prior and class-conditional probabilities

08/12/2022

EC-350 AI and DSS

EME (NUST)

38

38

Posterior Probability

Posterior, likelihood, evidence

- Using Bayes formula

$$P(w_j|x) = \frac{P(x|w_j) P(\omega_j)}{P(x)}$$

where in case of two categories

$$P(x) = \sum_{j=1}^2 P(x | \omega_j) P(\omega_j)$$

- Posterior = (Likelihood $\times$ Prior) / Evidence
- Evidence $P(x)$ can be viewed as a scale factor that guarantees that the posterior probabilities sum to 1
- $P(x|\omega_j)$ is called the likelihood of ω_j with respect to x ; the category ω_j for which $P(x|\omega_j)$ is large is more likely to be the true category

08/12/2022

EC-350 AI and DSS

EME (NUST)

39

39

Posterior Probability

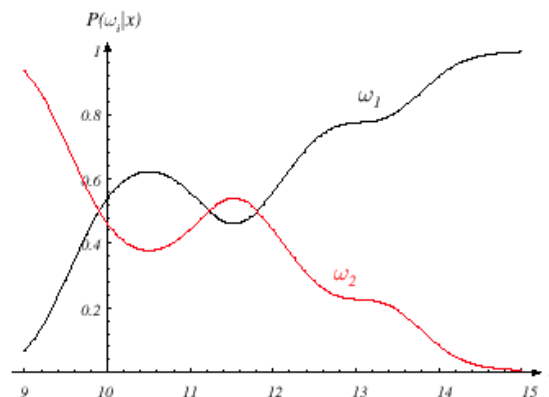


FIGURE 2.2. Posterior probabilities for the particular priors $P(\omega_1) = 2/3$ and $P(\omega_2) = 1/3$ for the class-conditional probability densities shown in Fig. 2.1. Thus in this case, given that a pattern is measured to have feature value $x = 14$, the probability it is in category ω_2 is roughly 0.08, and that it is in ω_1 is 0.92. At every x , the posteriors sum to 1.0. From: Richard O. Duda, Peter E. Hart, and David G. Stork, *Pattern Classification*. Copyright © 2001 by John Wiley & Sons, Inc.

08/12/2022

EC-350 AI and DSS

EME (NUST)

40

40